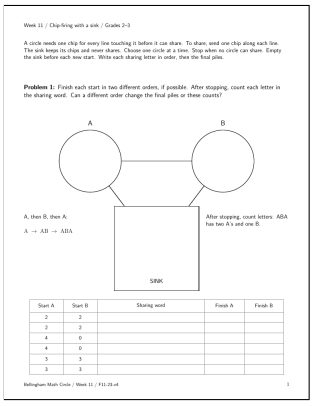
Week 11 / Chip-firing with a sink / Problem menu

for adults: pick, then print


Week 11 / Chip-firing with a sink / Grades 2–3

A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Finish each start in two different orders, if possible. After stopping, count each letter in the sharing word. Can a different order change the final piles or these counts?

A

B

A, then B, then A:

$A \to AB \to ABA$

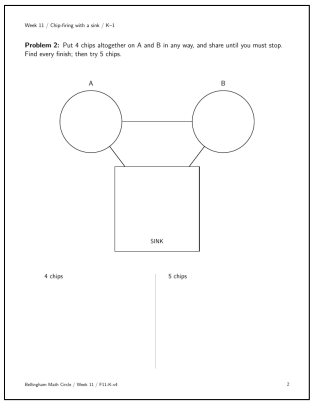
After stopping, count letters: ABA has two A's and one B.

SINK

Start A	Start B	Sharing word	Finish A	Finish B
2	2			
2	2			
4	0			
4	0			
3	3			
3	3			

Bellingham Math Circle / Week 11 / F11-23.v4 1

Two orders from each start on the triangle
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route
wording differs by band


Week 11 / Chip-firing with a sink / K–1

Problem 2: Put 4 chips altogether on A and B in any way, and share until you must stop. Find every finish; then try 5 chips.

A

B

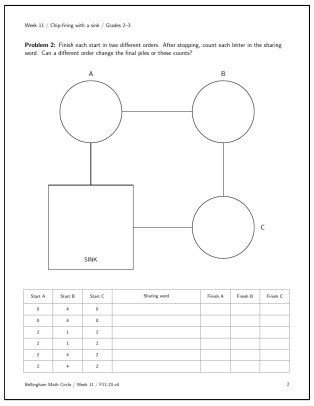
SINK

4 chips

5 chips

Bellingham Math Circle / Week 11 / F11-K.v4 2

Every finish from 4 chips, then from 5
K–1 P2
first route


Week 11 / Chip-firing with a sink / Grades 2–3

Problem 2: Finish each start in two different orders. After stopping, count each letter in the sharing word. Can a different order change the final piles or these counts?

A

B

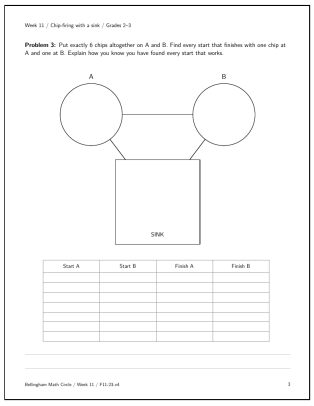
C

SINK

Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
0	4	0				
0	4	0				
2	1	2				
2	1	2				
2	4	2				
2	4	2				

Bellingham Math Circle / Week 11 / F11-23.v4 2

Two orders from each start on the four-cycle
K–1 P3 · Grades 2–3 P2
first route
wording differs by band


Week 11 / Chip-firing with a sink / Grades 2–3

Problem 3: Put exactly 6 chips altogether on A and B. Find every start that finishes with one chip at A and one at B. Explain how you know you have found every start that works.

A

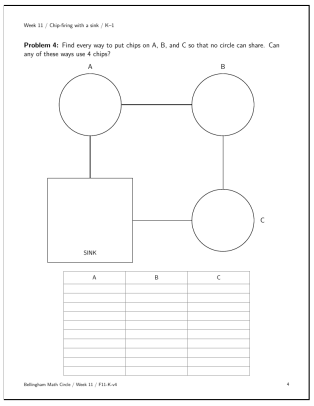
B

SINK

Start A	Start B	Finish A	Finish B

Bellingham Math Circle / Week 11 / F11-23.v4 3

Six-chip starts that finish at one and one
Grades 2–3 P3
first route


Week 11 / Chip-firing with a sink / K–1

Problem 4: Find every way to put chips on A, B, and C so that no circle can share. Can any of these ways use 4 chips?

A

B

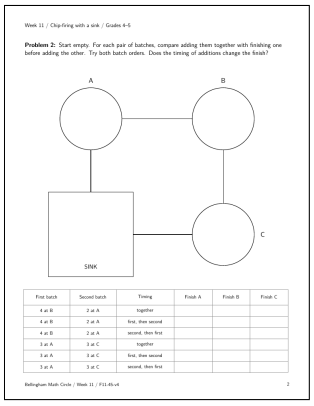
C

SINK

A	B	C

Bellingham Math Circle / Week 11 / F11-K.v4 4

Every placement where no circle can share
K–1 P4
return route


Week 11 / Chip-firing with a sink / Grades 4–5

Problem 2: Start empty. For each pair of batches, compare adding them together with finishing one before adding the other. Try both batch orders. Does the timing of additions change the finish?

A

B

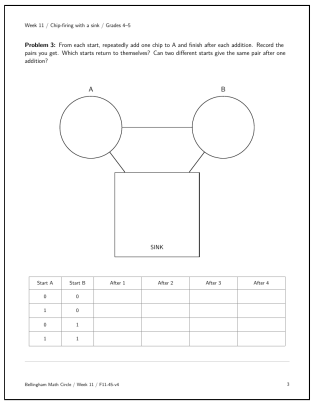
C

SINK

First batch	Second batch	Timing	Finish A	Finish B	Finish C
4 at A	2 at A	together			
4 at B	2 at A	first, then second			
4 at B	2 at A	second, then first			
3 at A	3 at C	together			
3 at A	3 at C	first, then second			
3 at A	3 at C	second, then first			

Bellingham Math Circle / Week 11 / F11-45.v4 2

Batches together or one at a time
Grades 4–5 P2
first route


Week 11 / Chip-firing with a sink / Grades 4–5

Problem 3: From each start, repeatedly add one chip to A and finish after each addition. Record the pairs you get. Which starts return to themselves? Can two different starts give the same pair after one addition?

A

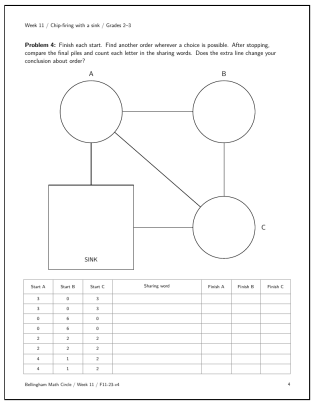
B

SINK

Start A	Start B	After 1	After 2	After 3	After 4
0	0				
1	0				
0	1				
1	1				

Bellingham Math Circle / Week 11 / F11-45.v4 3

Return and merging when adding at A
Grades 4–5 P3
first route


Week 11 / Chip-firing with a sink / Grades 2–3

Problem 4: Finish each start. Find another order wherever a choice is possible. After stopping, compare the final piles and count each letter in the sharing words. Does the extra line change your conclusion about order?

A

B

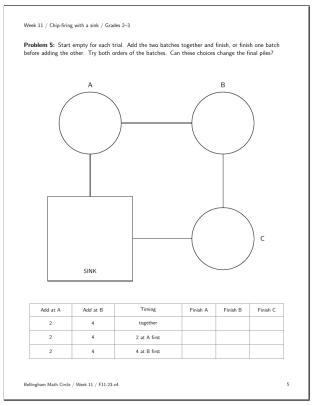
C

SINK

Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
3	0	3				
3	0	3				
0	6	0				
0	6	0				
2	2	2				
2	2	2				
4	1	2				
4	1	2				

Bellingham Math Circle / Week 11 / F11-23.v4 4

Order on the board with an extra line
Grades 2–3 P4 · Grades 4–5 P4
return route
wording differs by band


Week 11 / Chip-firing with a sink / Grades 2–3

Problem 5: Start empty for each trial. Add the two batches together and finish, or finish one batch before adding the other. Try both orders of the batches. Can these choices change the final piles?

A

B

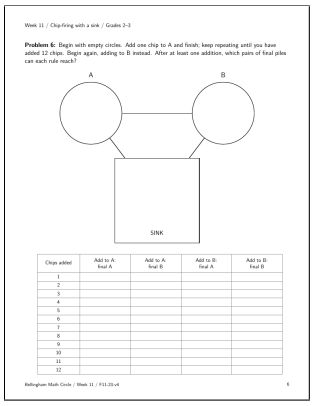
C

SINK

Add at A	Add at B	Timing	Finish A	Finish B	Finish C
2	4	together			
2	4	2 at A first			
2	4	4 at B first			

Bellingham Math Circle / Week 11 / F11-23.v4 5

Adding chips while sharing, or in batches
K–1 P5 · Grades 2–3 P5
return route
wording differs by band


Week 11 / Chip-firing with a sink / Grades 2–3

Problem 6: Begin with empty circles. Add one chip to A and finish; keep repeating until you have added 12 chips. Begin again, adding to B instead. After at least one addition, which pairs of final piles can each rule reach?

A

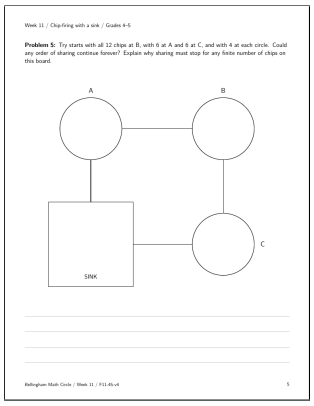
B

SINK

Chips added	Add to A: final A	Add to A: final B	Add to B: final A	Add to B: final B
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				

Bellingham Math Circle / Week 11 / F11-23.v4 6

Adding one chip at a time on the triangle
K–1 P6 · Grades 2–3 P6
return route
wording differs by band


Week 11 / Chip-firing with a sink / Grades 4–5

Problem 5: Try starts with all 12 chips at B, with 6 at A and 6 at C, and with 4 at each circle. Could any order of sharing continue forever? Explain why sharing must stop for any finite number of chips on this board.

A

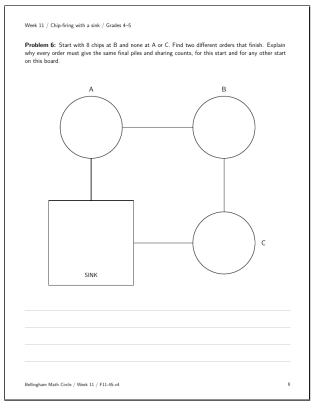
B

C

SINK

Bellingham Math Circle / Week 11 / F11-45.v4 5

Why sharing must stop on the four-cycle
Grades 4–5 P5
return route


Week 11 / Chip-firing with a sink / Grades 4–5

Problem 6: Start with 8 chips at B and none at A or C. Find two different orders that finish. Explain why every order must give the same final piles and sharing counts; for this start and for any other start on this board.

A

B

C

SINK

Bellingham Math Circle / Week 11 / F11-45.v4 6

Why every order gives the same piles and counts
Grades 4–5 P6
return route